

PENGZHEN LI (李鹏振)

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EDUCATION

University of Tennessee, Knoxville
Ph.D. in Economics

expected 2027

Shandong University, Jinan

2018

M.A. in Finance — thesis published in *Technological Forecasting and Social Change* (2019); 1,000+ citations

Tongji University, Shanghai

2015

B.A. in Economics

RESEARCH INTERESTS

International Trade, Environmental and Energy Economics, Applied Econometrics, and Political Economy.

PEER-REVIEWED PUBLICATIONS

Four SSCI/SCI-indexed publications; 1,132 total citations, h-index 4 (Google Scholar, August 2026).

“Do green technology innovations contribute to carbon dioxide emission reduction? Empirical evidence from patent data.” K. Du, **P. Li**, and Z. Yan. *Technological Forecasting and Social Change*, 146 (2019), 297–303. **1,000+ citations (Google Scholar)**.

“Spatial optimization of the sustainable aviation fuel supply chains from forest residues via fast pyrolysis/hydrotreatment considering feedstock ash content variability.” **P. Li**, T. E. Yu, N. Labbé, N. Abdoulmoumine, M. Garcia-Perez, K. P. Hoyt, and B. C. English. *Biomass and Bioenergy*, 208 (2026), 108793.

“Assessing the impact of preprocessing and conversion technologies on the sustainable aviation fuel supply from forest residues in the Southeast USA.” **P. Li**, T. E. Yu, C. Trejo-Pech, J. A. Larson, B. C. English, and D. N. Lanning. *Transportation Research Record*, 2678(9) (2024), 639–654.

“Beef price spread relationship with processing capacity utilization.” C. Martinez, **P. Li**, C. N. Boyer, T. E. Yu, and J. G. Maples. *Journal of the Agricultural and Applied Economics Association*, 2(1) (2023), 133–145.

WORKING PAPERS

Measured Quality Falls Within the Firm: Market Access and Chinese Exports with James Lake [**Job Market Paper**] — evidence from Chinese customs records, 2000–2014. [[paper](#)]

Abstract: This paper links the 2002 removal of tariff uncertainty on Chinese exports to the United States to a decline in measured quality within continuing export relationships. Comparing a firm’s US and non-US shipments of the same product, measured quality falls by about 6 percent among cells that ship in both periods in the three years before an overlapping quota removal, and by about 11 percent in the full sample over the whole window, which contains that removal. Restricting the sample to cells that ship in both periods retains 83 percent of the full-window estimate and yields a decline twice as large as the full-sample estimate in the three clean years. The price and quantity responses reject a markup cut on an unchanged demand schedule. Measured quality is a demand

residual built from price and quantity, so it does not distinguish physical downgrading from a cheaper mix of varieties. We develop a model in which improved access can extend a seller's reach to buyers below a sorting boundary, who lower the return to appeal. Under stated conditions, appeal is capped and can fall. The continuing-cell price and quantity pattern is consistent with this mechanism but does not identify it against other sources of a demand-residual shift.

Trade Restrictiveness Indices with Discriminatory Tariffs with James Lake. *Presented at the Midwest Economic Theory and International Trade Meetings, Virginia Tech, 2025.* [\[paper\]](#)

Abstract:

A Mean-Field Game of Mayoral Endorsements: Identification and Evidence from Brazilian Coalitional Presidentialism with Matt Van Essen and Luiz Lima. *2024 J. Fred and Wilma Holly Award (best second-year paper).* [\[paper\]](#) [\[slides\]](#)

Abstract:

Price Risk, Irreversible Investment, and the Welfare Benchmark for the Section 45Z Clean Fuel Production Credit with T. Edward Yu. [\[paper\]](#) [\[slides\]](#)

How Should 45Z Pay for Risk? A Welfare-CVaR and Project-Finance Decomposition of the IRA Clean Fuel Production Credit with T. Edward Yu. [\[paper\]](#) [\[slides\]](#)

Estimating Pork Primal Price Relationships Using a Threshold Vector Autoregressive Analysis with Charles Martinez, Christopher N. Boyer, T. Edward Yu, and David Anderson. *SSRN working paper 4121953 (2022).* [\[paper\]](#)

WORK IN PROGRESS

A Weight-Free Benchmark for Multilateral Tariff Bargaining with Georg Schaur. *Draft, August 2026.* [\[paper\]](#) [\[slides\]](#)

PRESENTATIONS

Midwest Economic Theory and International Trade Meetings, Virginia Tech	Spring 2025
University of Tennessee, Economics Brown Bag Seminar Series	Spring 2024, Fall 2024
University of Tennessee, Applied Micro Group	Spring 2024, Fall 2024
Western Regional Science Association, Annual Meeting, Hawaii	2023
Western Agricultural Economics Association, Annual Meeting, Santa Fe, NM	2022
Southern Agricultural Economics Association, Annual Meeting, New Orleans, LA	2022

PROFESSIONAL EXPERIENCE

University of Tennessee – Knoxville, TN 2021 – present
Graduate Research & Teaching Assistant. USDA NIFA and FAA ASCENT projects (2021–2023): mixed-integer supply-chain optimization (GAMS), IMPLAN/DEA siting analysis, and spatial data pipelines (ArcGIS, R, Python) for forest-residue sustainable aviation fuel; two publications and three working papers.

Shandong University – Jinan, China 2016 – 2018
Graduate Research Assistant. Panel-threshold and machine-learning analysis of green innovation and CO₂ emissions (published in *TFSC*, 2019).

TEACHING EXPERIENCE

University of Tennessee

Instructor

ECON 351: Monetary Economics	Spring 2025, Spring 2026
ECON 322: The Global Economy: Trade and Development	Summer 2025
ECON 313: Intermediate Macroeconomics	Summer 2026

Teaching Assistant

Ph.D. courses:

ECON 511: Microeconomic Theory	Fall 2023, Fall 2024, Fall 2025
ECON 512: Advanced Microeconomics	Spring 2024
ECON 581: Mathematical Methods in Economics	Summer 2024

Master's courses:

AREC 525: Agribusiness Operations Research Methods	Spring 2022, Spring 2023
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HONORS AND AWARDS

Charles B. Garrison Award for Excellence in Teaching, University of Tennessee	2026
Graduate Summer Fellowship, University of Tennessee	2025
J. Fred and Wilma Holly Award (best second-year paper), University of Tennessee	2024
J. Fred and Wilma Holly Fellowship, University of Tennessee	2023 – present
Graduate Assistantship, University of Tennessee	2021 – present
Outstanding Rainbow Volunteer, Tongji University (tutoring pupils in Shanghai)	2013 – 2015

SKILLS

Programming Languages: Stata, R, Python, SQL, GAMS, MATLAB, LaTeX, ArcGIS, IMPLAN.

Methods: panel and causal econometrics (FE, IV, DiD, event study, SCM, GMM, panel threshold), time series (TVAR, VECM, mGARCH), mixed-integer and stochastic programming, DEA, cooperative and mean-field games, machine learning.

Credentials: CFA Level II.

Languages: Chinese (native), English (fluent).

REFERENCES

James Lake (Chair)

Associate Professor of Economics

Department of Economics

University of Tennessee

Email: jlake7@utk.edu

Georg Schaur

Leigh Burch III Economics Professor and Haslam Family Faculty Fellow

Department of Economics

University of Tennessee

Email: gschaur@utk.edu

T. Edward Yu

Professor of Agricultural and Resource Economics

Department of Agricultural and Resource Economics

University of Tennessee

Email: tyu1@tennessee.edu